

Study Information

1. **Title:** Communication selectivity in the STS
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3. **Description:**

Previous studies have reported selectivity in the posterior superior temporal sulcus (pSTS) to viewing other people engage in a social interaction (Isik et al., 2017; Lee Masson & Isik, 2021; Walbrin et al., 2018). A recent large-scale naturalistic analysis of STS responses found that this region does not respond to all kinds of social interactions (e.g., playing tennis, carrying a table, talking) but to communicative interactions, in particular (McMahon et al., 2023). However, that study was not designed to test for communicative selectivity and used a deep phenotyping approach in which a large amount of data was collected in a small number of participants. For these reasons, the generalizability of communicative selectivity is unknown. The current experiment is a controlled investigation of selectivity to communicative interactions between two people in the STS. We will also test generalizability of these representations to a) inferred third-party communicative interactions in which only one person is visible, talking but an unseen partner is inferred, and b) first-person interactions in which a single person is communicating with the viewer.

4. **Hypotheses:**

Our main hypothesis is that regions of the STS respond more to communicative than noncommunicative events. We will test this by defining functional regions of interest (fROIs) in the STS based on existing contrasts in the literature and determining whether these fROIs respond more to dyadic communicative events than dyadic independent or joint actions and individual faces communicating than not communicating (**Figure 1**). We will also test this in the whole brain using group random effects analyses. If our hypothesis is true, we should find one or more regions of the STS that respond more to both dyadic communication and individual faces communicating.

As a control to confirm that selectivity for communication is localized to the STS, we will also perform the same analyses in other regions of interest outside the STS including early visual cortex (EVC), motion-selective middle temporal area (MT), face-selective fusiform gyrus (FFA), extrastriate body area (EBA), and theory-of-mind-selective temporal parietal junction (TPJ). In each of these regions, we expect that there should not be greater response to dyadic communication compared to other dyadic events and to facial communication compared to facial noncommunication.

As a final control, we will test whether each fROI responds more to social conditions than inanimate object motion. To do this, we will compare the response in the fROIs to faces or dyadic action vs objects. For FFA and all STS fROIs, we expect a greater response to faces and dyads than objects. For EVC, MT, and TPJ, there should be no difference between social conditions and objects. Finally, EBA would be expected to respond more to dyads than objects but not faces more than objects.

As a secondary hypothesis, we will test whether any region is selective for first-person or third-person interactions. If such a region exists, we expect the region would respond differently to faces that appear to be communicating with the viewer than faces that



Figure 1. Example stimuli for each experimental conditions for the main experimental runs: face communicating to the viewer, faces communicating off screen, faces not communicating, objects, bodies (not shown), communication dyads (paired with independent videos), independent dyads, communication dyads (paired with joint action videos), and joint action dyads.

appear to be communicating with an unseen other person. We will look for this both within fROIs and using a group random-effects analysis.

Design Plan

5. **Study type:** Experiment
6. **Blinding:** No blinding involved
7. **Study design**

Overview

In the experiment, every subject will complete 1) a high-resolution anatomical scan, 2) two runs of a theory of mind localizer either a false belief task (Saxe & Kanwisher, 2003) or emotional pain task (Bruneau et al., 2015) depending on prior participation, and 3) nine runs of the main experiment.

Main experiment

Stimuli: The main experiment has a block design which will include 9 conditions: objects, bodies, faces communicating to the viewer, faces communicating to an unseen person off screen, faces performing noncommunicative actions, communication dyads (paired with independent videos), independent dyads, communication dyads (paired with joint action videos), and joint action dyads (Figure 1).

Each condition has 10 unique 2 s videos. Face and dyad videos were collected from stock video sites and were trimmed to 2 s. Face videos were additionally manually cropped to a square around one face in the video.

The dyad videos are paired so that for each video depicting a communicative interaction there is a paired video with the same people, in a similar scene context performing an independent or joint action.

Object and body stimuli were randomly selected from stimuli used to localize category selective visual areas (Julian et al., 2012). Body stimuli are included to localize extrastriate body area (EBA) but are not included in the main analyses. All stimuli have been pre-registered and are available on the [OSF project page](#).

Experimental procedure: In the main experiment, participants will perform a 1-back task. During each block of the experiment, there will be 5 unique videos, and one of the videos will randomly be selected to be repeated, and the participant should respond on that trial.

One run of the main experiment will consist of 18 stimuli blocks (2 repetitions of each of the 9 conditions). Every 3 blocks, there will be an 18 s fixation period. Each run will also begin and end with an 18 s fixation period.

The order of the conditions in the experiment will be randomly determined for each participant. Following this block order, within run 1 the conditions will be presented in a palindromic order (e.g., ABCCBA). The order of the conditions will rotate by one between runs so that across all 9 runs of the experiment, each condition will appear in every position (e.g., run 1: ABCCBA, run 2: BCAACB, run 3: CABBAC). The experiment presentation code is available on [GitHub](#).

8. **Randomization:** The assignment of videos to blocks, the order of blocks, the presentation of videos within a block, and the video repeated within a block will all be randomly determined per participant.

For each participant, the 10 videos per condition were randomly split into groups of 5 videos. For the dyad videos, this is done with respect to the pairs. Paired stimuli are not presented in the same run to maximize visual variability within run. In the first half of the experiment (runs 1-5), the participants view one half of the videos per condition, and in the second half of the experiment (runs 6-9), the other half of the videos are presented. The code determining randomization and block order is available on [GitHub](#).

Sampling Plan

9. **Existing data:** Registration prior to analysis of the data.
10. **Explanation of existing data:** Data collection and fMRI preprocessing are in progress during preregistration. Preprocessing is only performed as a check of data quality and no analyses or summary statistics related to the research question have been performed.
11. **Data collection procedures:** Participants will be recruited from the MIT community and local Cambridge area through email, flyers, or word-of-mouth. They will have no known MR contraindications and normal or corrected-to-normal vision. Participants will be compensated \$75 for their time.
12. **Sample size:** Our target sample size is 14 participants (see Sample size rationale). We may need to recruit additional participants to reach our inclusion criteria (see Exclusion criteria). If key effects are of borderline significance in our first group of 14 participants, we will run an additional 6 participants.
13. **Sample size rationale:** Based on pilot analyses in two participants, we conducted power analyses between conditions of interest. We found that 14 participants would be

needed to achieve significance at a power of 0.95 and an alpha level of 0.05 for a contrast of the “face first” and “face noncommunicative” conditions in a fROI that responds more to communication than independent or joint actions.

We do not have a sample size estimate for a whole-brain random effects analysis. If a whole-brain random effects contrast for communication > (independent and joint action) is marginal at 14 participants, we will recruit an additional 6 participants.

14. **Stopping rule:** We may terminate the study early if there are not consistent patterns of activation within the first 5 participants.

Variables

15. **Manipulated variables:** For the face conditions, we manipulate whether the faces are communicating (face – communicating to viewer vs face – not communicating, face – communicating off screen vs face – not communicating) and whether they appear to be communicating to the participant or an unseen other person (face – communicating to viewer vs face – communicating off screen).

The dyad videos manipulate whether two people are communicating to one another relative to performing an independent action (communicating vs independent) or are communicating relative to performing a joint action (communicative vs joint action).

Object stimuli are presented as a nonsocial control condition.

The stimuli variation across exemplars is intended to control for low-level visual effects.

16. **Measured variables:** The key measured variable is the fMRI response (beta-estimate). We may also use behavioral performance as an inclusion criterion.

Analysis Plan

17. **Statistical models**

fMRI preprocessing:

fMRI data will be preprocessed using standard preprocessing using fMRIPrep 24.1.1. Data will be normalized to the MNI152NLin2009cAsym template (Fonov et al., 2009, 2011) to enable group-level random-effects analyses. Preprocessing will follow default settings except for BOLD to Anatomical parameters which may be modified per subject to achieve good alignment.

To get the response estimates for each condition per run, we will run a GLM analysis using Nilearn. Data will be smoothed using a 5 mm full-width half-max kernel. Motion will be corrected with 6 parameter translation and rotation parameters, and time points of greater than 0.5 mm framewise displacement and segments shorter than 5 frames will be scrubbed.

fROI definition

We will define fROIs in our data in one of two ways: either using separate functional localizer scans or using held-out data in the main experimental runs.

Our fROIs defined in separate functional localizer scans will be the TPJ and SI-STS. For TPJ, we will use data from either the false belief (Dodell-Feder et al., 2011) or emotional pain task (Bruneau et al., 2015) and then take 10% most selective voxels in a TPJ parcel from (Dufour et al., 2013). Similarly, we will localize SI-pSTS using a prior task (Isik et al., 2017) and will take the 5% most selective voxels in an STS parcel (Deen et al., 2015).

For all other regions, we will localize the fROI using all but 1 run of the main task. We will use the held out run to estimate the condition response in the fROI and do this iteratively to leave out each run once. For EVC and MT, we will get the response for all conditions relative to fixation and then take the 10% most responsive voxels in EVC (V1d, V1v, V2d, V2v) or MT from (Wang et al., 2014). For EBA (bodies minus objects), FFA (all face conditions minus objects), and fSTS (all face conditions minus objects), we will take the 10% most selective voxels in a parcel from (Julian et al., 2012).

We will also localize our own fROIs by contrasting the face communicating conditions from the noncommunicating condition at taking the 5% most selective voxels in an STS parcel from (Deen et al., 2015). Using the same parcel and proportion, we will define an fROI that responds more to communicating dyads than independent dyads and one that responds more to communicating dyad than joint actions.

fROI analysis

Within each fROI, we will estimate the response to every condition (except bodies) on the run that was held out from defining the fROI. We will do this iteratively for every run and then average across runs within participant.

To determine if an fROI responds more to dyadic communication either than to dyadic independent actions or joint actions, we will compute paired-sample, one-tailed t-tests between dyadic communication and each control condition.

We will also test whether as fROI responds more to faces communicating than not communicating. For each face communicating condition, we will compute a paired-sample, one-tailed t-test against faces not communicating. We will correct these two tests for multiple comparisons using false-discovery rate (FDR) correction at a level of $\alpha = 0.05$.

We will test whether there are greater responses to faces and dyads than objects in each ROI by computing paired-sample t-tests between the average response to faces or dyads and objects. These tests will again be FDR corrected.

To test our secondary hypothesis, that there will be a differential response to faces appearing to communicate with the viewer versus an unseen other person, we will compare the two faces communicating condition using a paired-sample, two-tailed t-test in each fROI.

Whole-brain random effects analysis

We will also compute a whole-brain random effects analysis in Nilearn for a contrast of each (1) communicating vs noncommunicating faces, (2) dyadic communication vs dyadic independent actions, (3) dyadic communication vs dyadic independent actions, and (4) faces communicating to the viewer versus off-screen.

18. Inference criteria:

We will infer whether a result is present in the fROI analysis based on p-values at an α -level of 0.05. For the random effects analysis, in the most liberal case, we will infer greater response to a given condition at uncorrected $p < 0.01$.

19. Data exclusion:

We will exclude fMRI runs with excessive motion artifacts. Motion artifacts will be determined using the standardized framewise displacement (FD) from fMRIPrep. We will scrub fMRI frames with greater than 0.5 mm FD and remove segments of fewer than 5 frames after motion scrubbing based on defaults in Nilearn 0.13.0. If a run has more than 12 frames removed (corresponding to the duration of 2 experimental blocks), that run will be removed from additional analyses.

We will also exclude runs with poor behavioral performance indicating that participant was inattentive or asleep. For instance, if participants respond incorrectly on 50% or more repeated trials, those runs may be removed.

Participants with 3 or more runs excluded due to either motion artifacts or behavioral performance will be excluded.

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